SOOHYUN CHOI

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
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RESEARCH INTERESTS

Reinforcement learning and sequential decision making, with emphasis on multi-step policy improvement, analysis of tau regimes, offline goal-conditioned RL, and generative long-horizon state-space planning.

EDUCATION

Hanyang University 2024-Present
Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong

Hanyang University 2018-2024
B.S. in Electronic Engineering

MANUSCRIPT & SELECTED RESEARCH

S. Choi, S. Cho, and S. Hong, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning." *Manuscript, 2026.*

- Developed explicit state-space bridges, inverse-dynamics action chunks, and execution-time replanning for long-horizon offline GCRL; evaluated on OGBench navigation and manipulation tasks.

S. Choi, M. Oh, and I. Song, "ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold." *Undergraduate Capstone Technical Report, 2024; revised 2026.* [PDF]

- Formulated designer-selected performance equivalence classes for circuit search and combined the induced quotient geometry with attention-weighted line search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Multi-step Proximal Policy Improvement 2026-Present

- Analyze how proximal step size tau governs recursively re-centered policy refinement, using sweeps and mathematical displacement diagnostics to characterize local-flow, nonlocal finite-step, and over-refinement regimes.

PathFlower - Goal-Conditioned State Flows Current

- Extend PathBridger from an explicit subgoal and bridge toward final-goal-conditioned state flows that generate useful intermediate states for long-horizon planning.

REACT: Recursive Algorithm for RC Network Transfer Functions 2024-2025
First-author industry-sponsored research · SK hynix-Hanyang University project · Unpublished manuscript

- Developed a recursive closed-form transfer-function method for arbitrary RC trees and a dominant-pole reduced-order approximation validated against SPICE.

HONORS & FUNDING

- Outstanding Graduation Project Award - KRW 2,000,000 award
- Excellence Award, College of Engineering Capstone Design - KRW 1,000,000 award
- Graduation Honors, Hanyang University - 2024
- BK Research Assistantship (BK-RA), Hanyang University - 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory

Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I